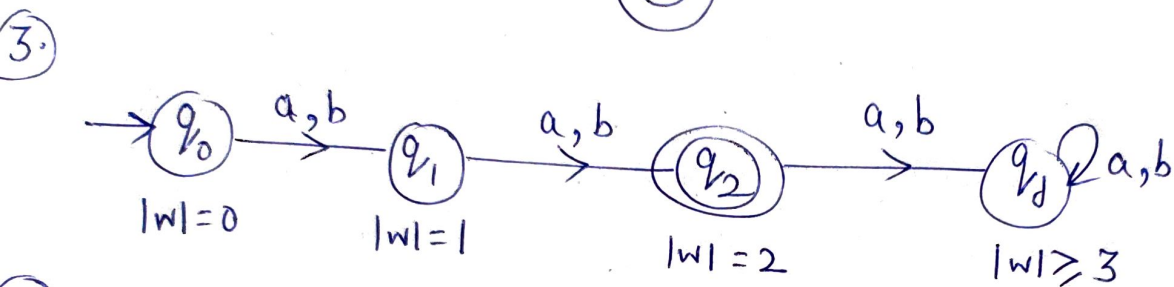
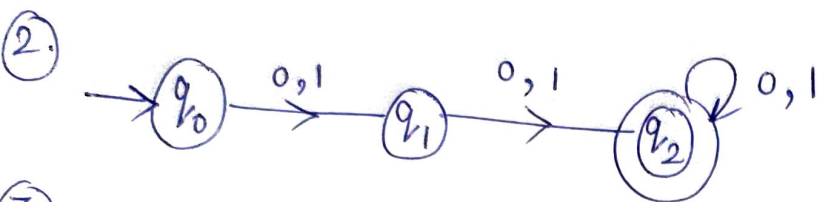
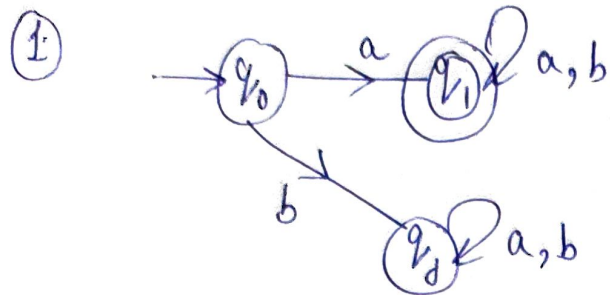


Tutorial sheet (TAFL)



④

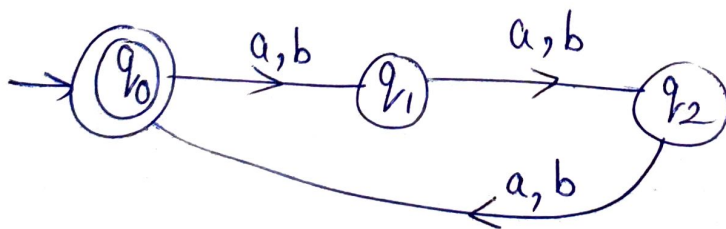
$L: |w| < 2$
 $L = \{ \epsilon, a, b \}$



⑤

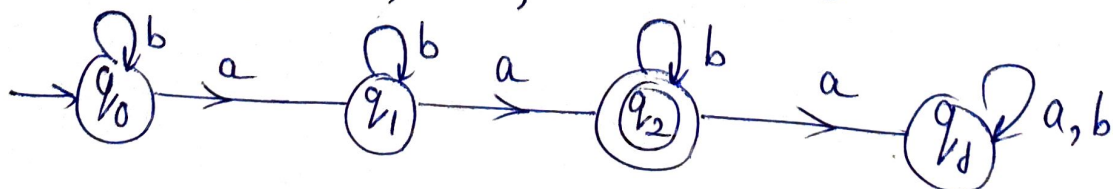
$L: |w| \bmod 3 = 0$
 $L = \{ \epsilon, aaa, aab, \dots \}$

It should be non-final state. Doesn't matter you take q_d or q_2 .

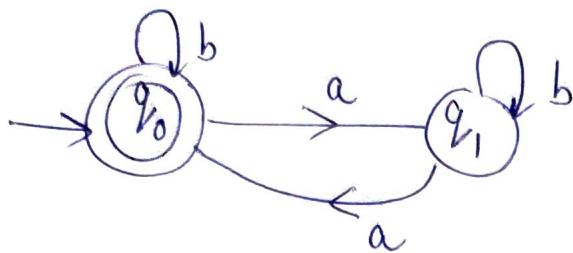


⑥

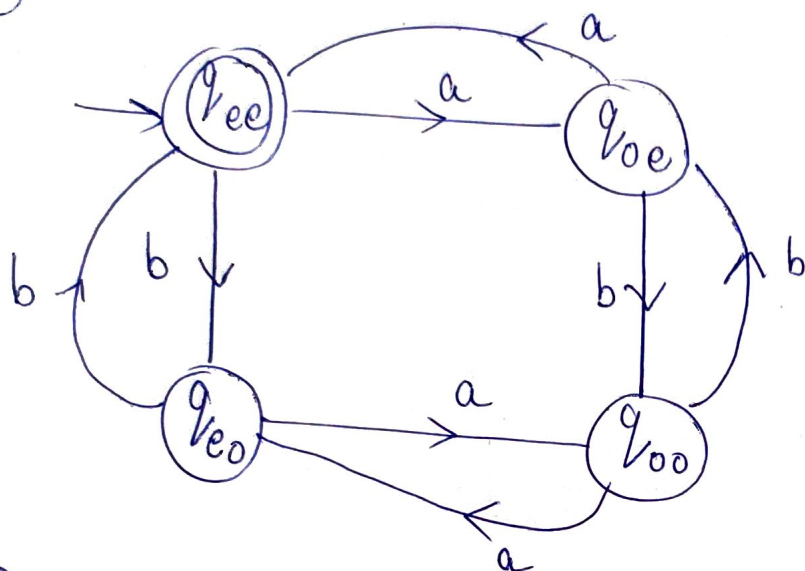
$L: n_a(w) = 2$
 $L = \{ aa, aab, aba, \dots \}$



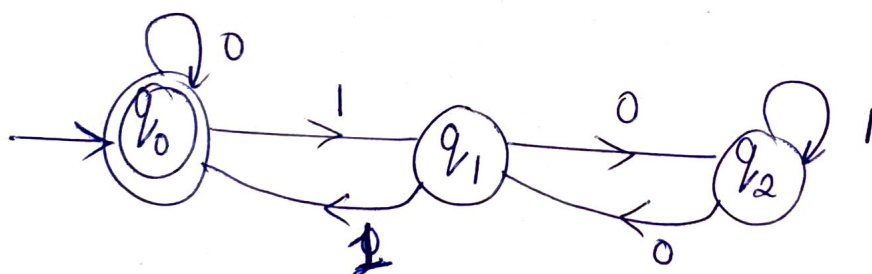
7. $L = \{ \epsilon, b, aa, \dots \}$



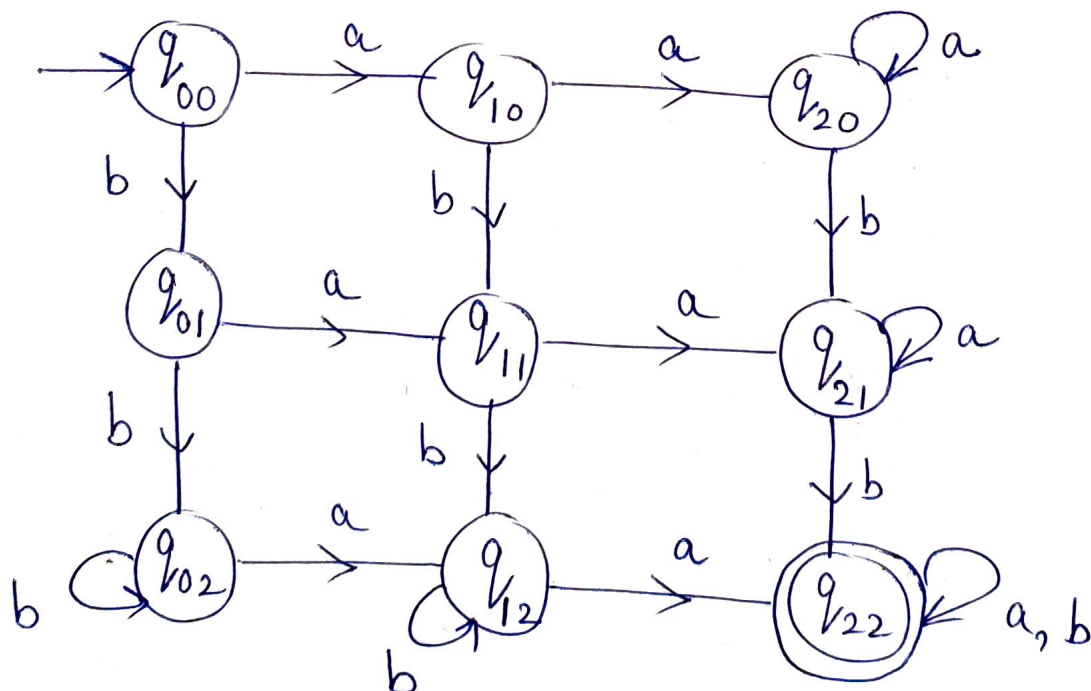
8.



9.



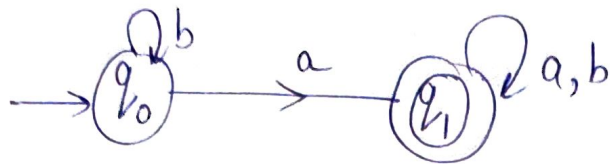
10.



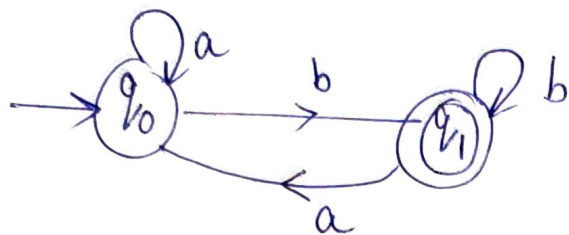
State Representation

q_{ij} represents no. of a's j rep no. of b's

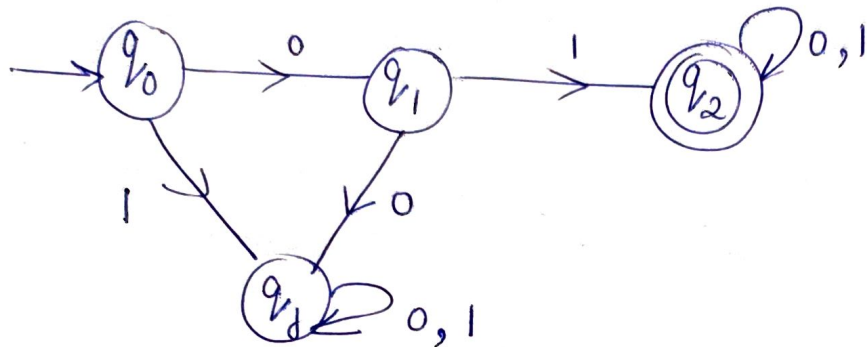
11.



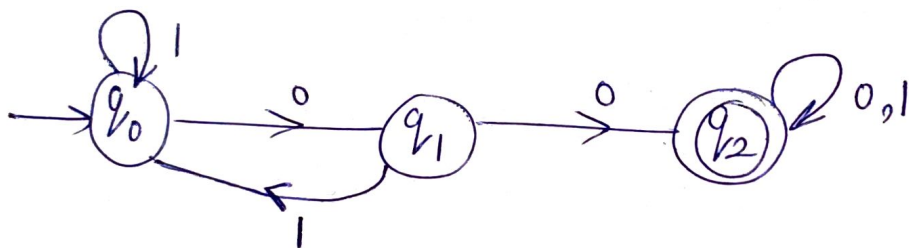
12.



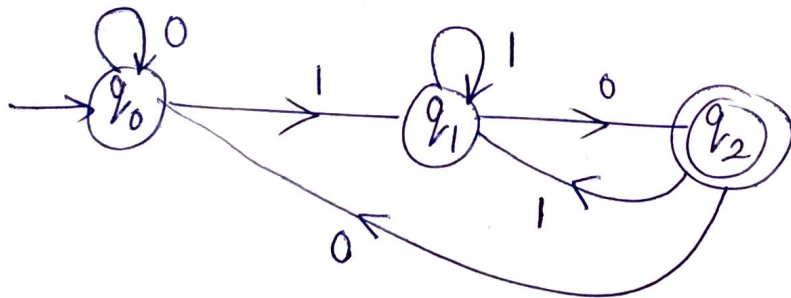
13.



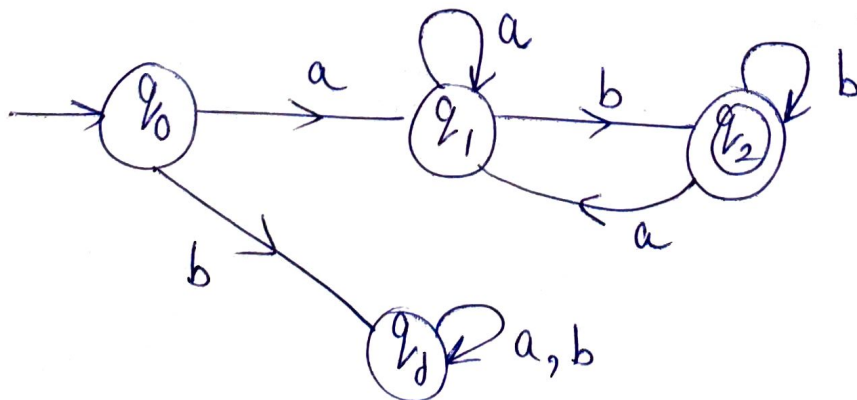
14.



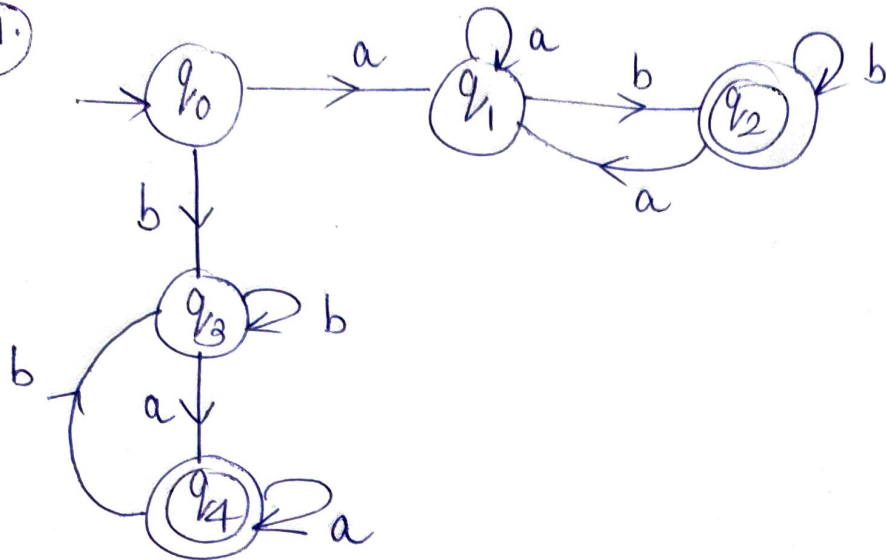
15.



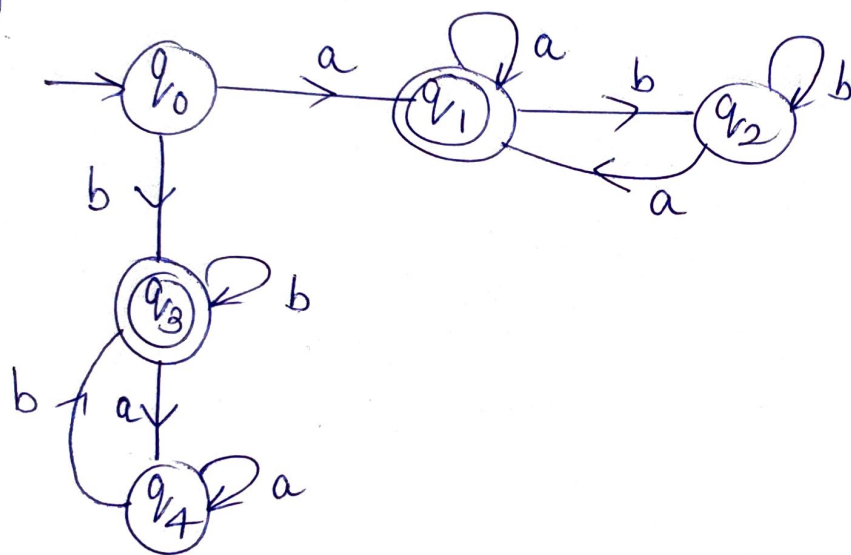
16.



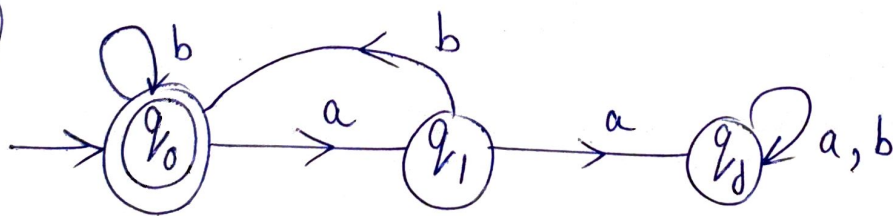
17.



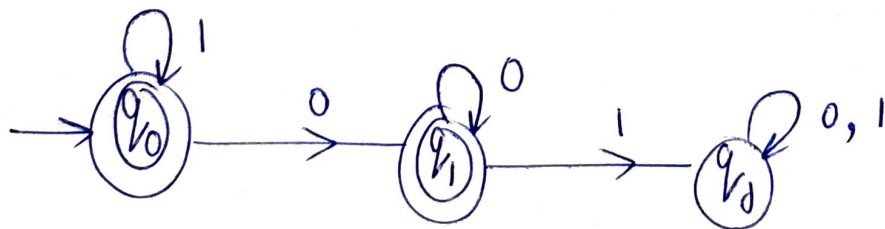
18.



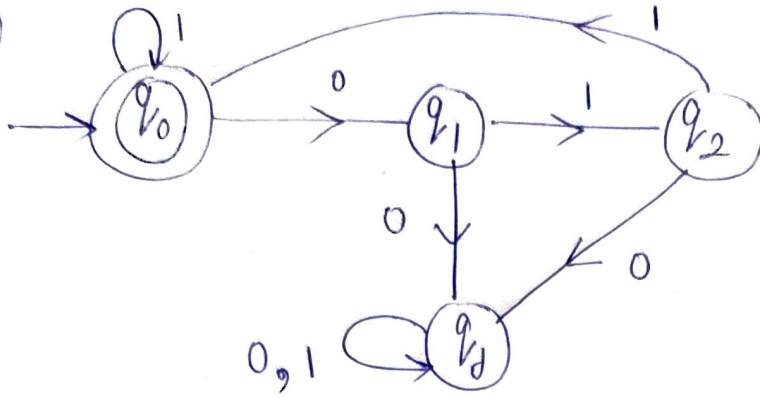
19.



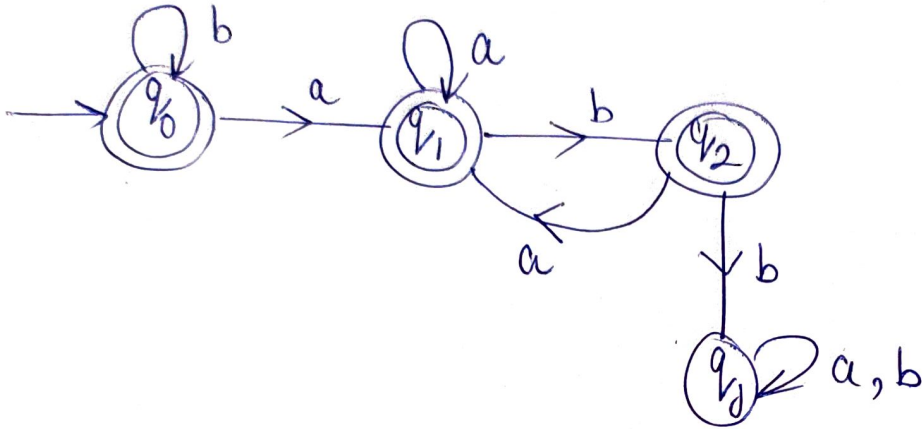
20.



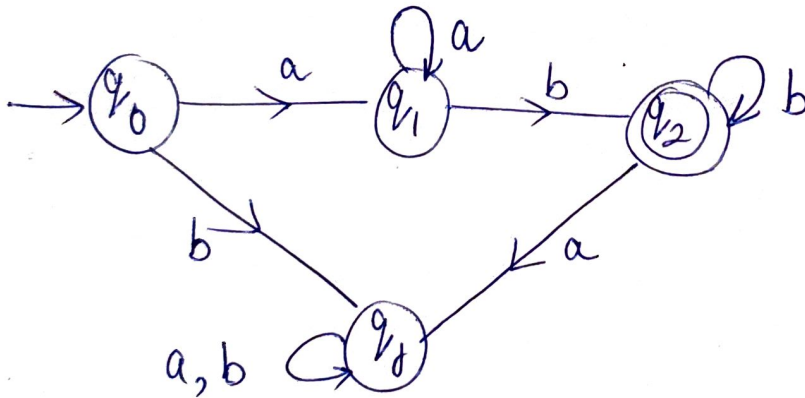
21.



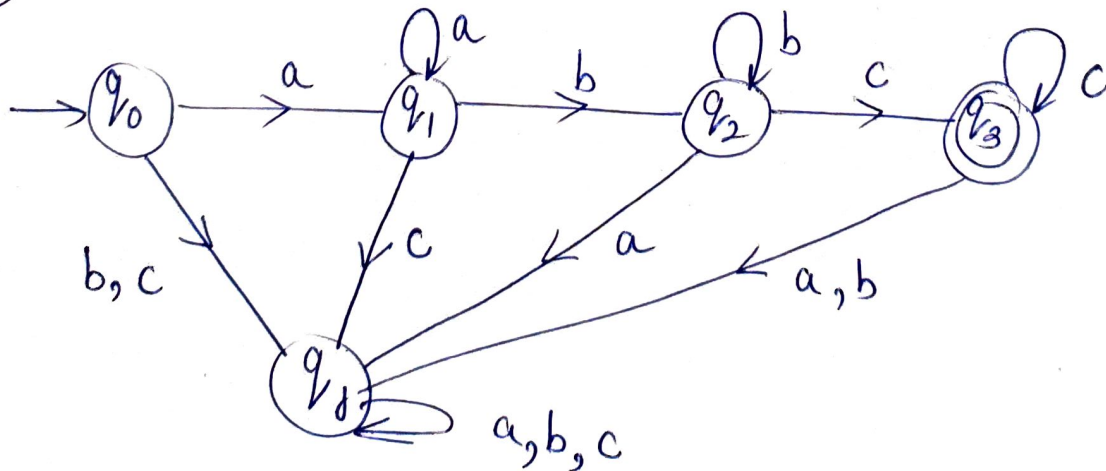
22.



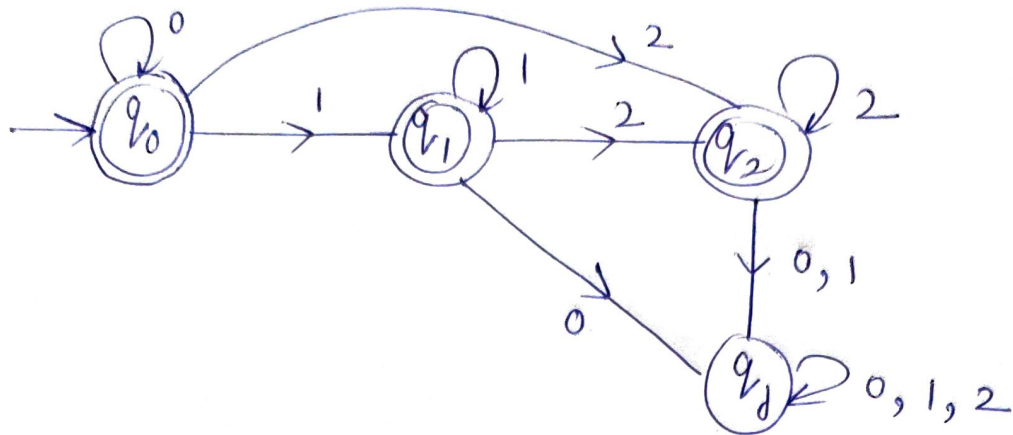
23.



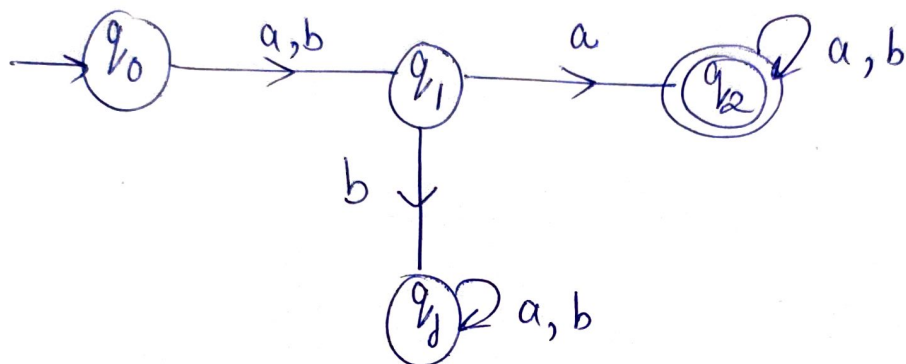
24.



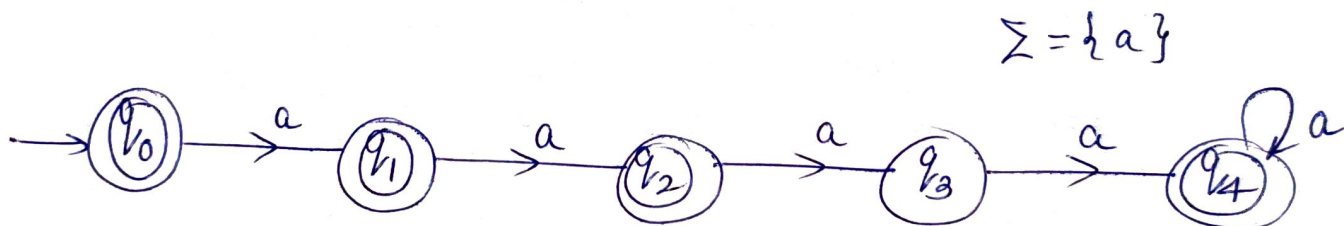
25.



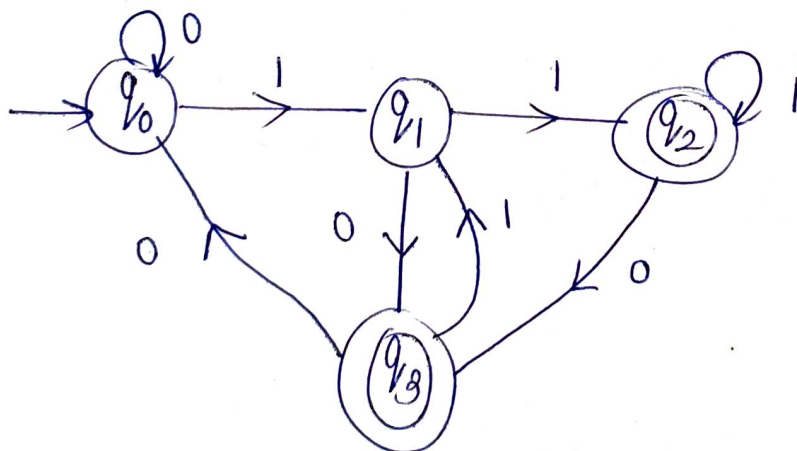
26.



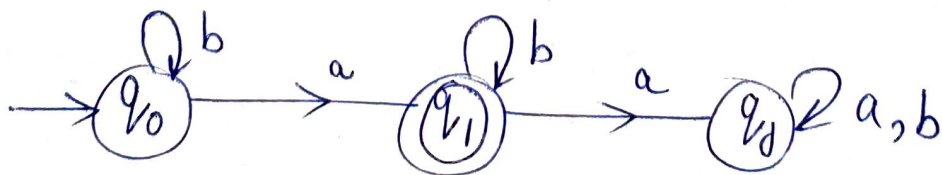
27.



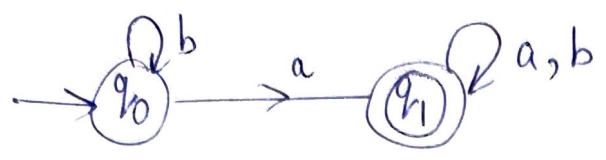
28.



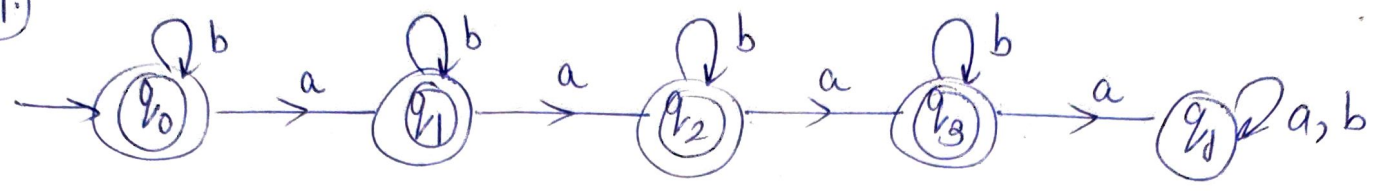
29.



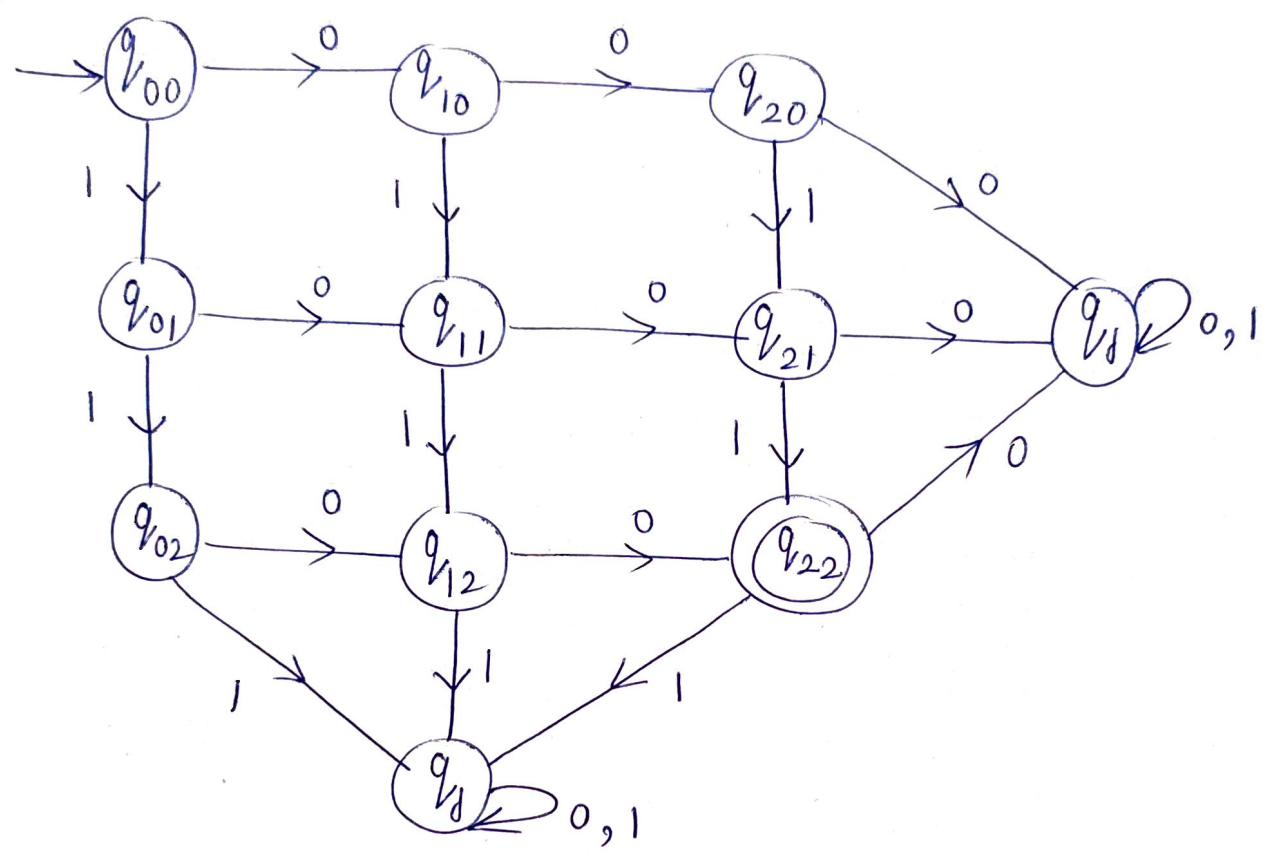
30.



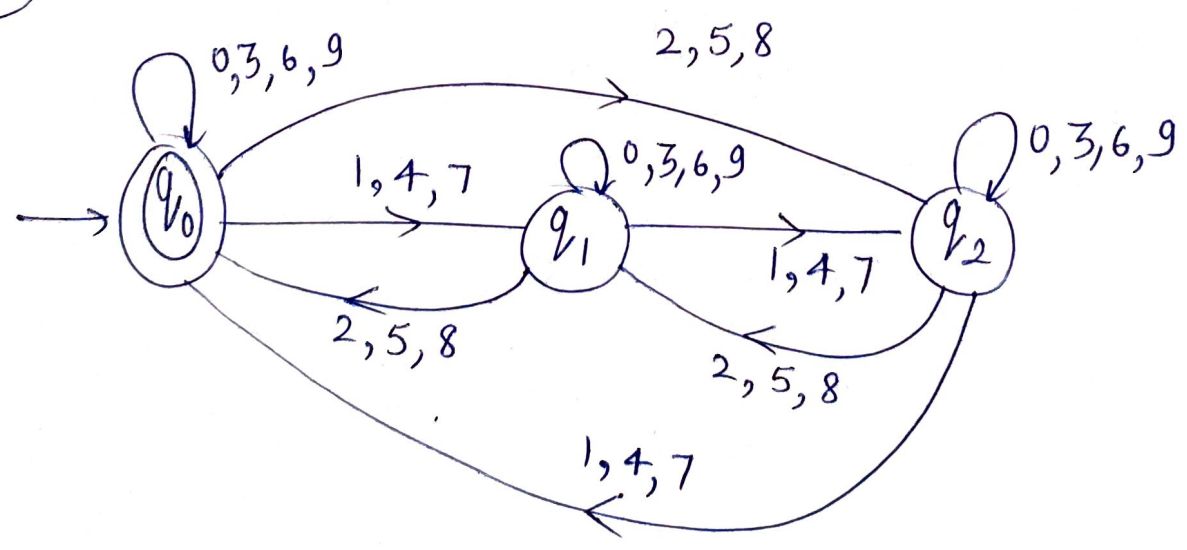
31.



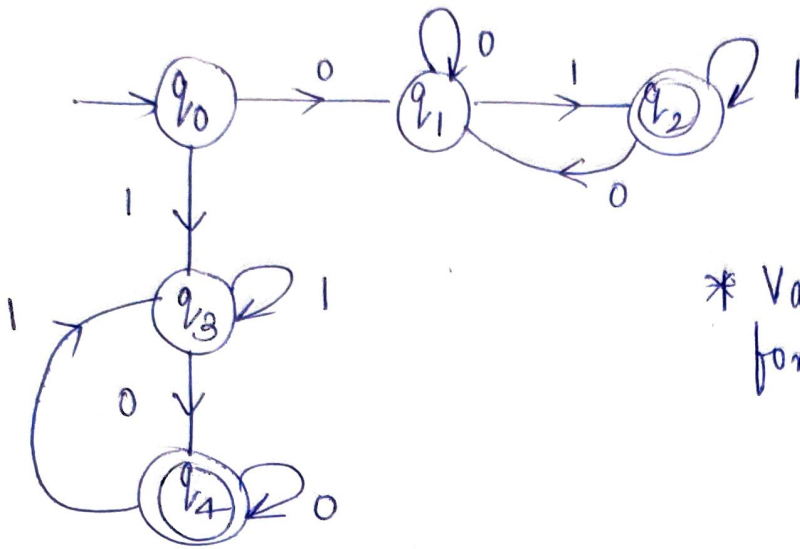
32.



33.



34. $L = \{ 01, 10, 001, 011, 100, 110, \dots \}$



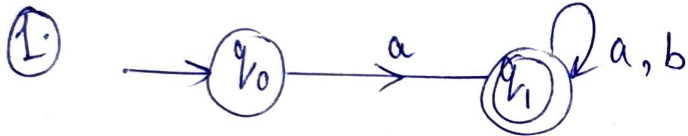
* Valid format: $\underline{0} \dots \checkmark$
or

$\dots \underline{0} \checkmark$

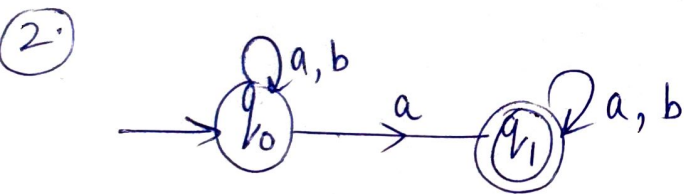
But

$\underline{0} \dots \underline{0} \times$

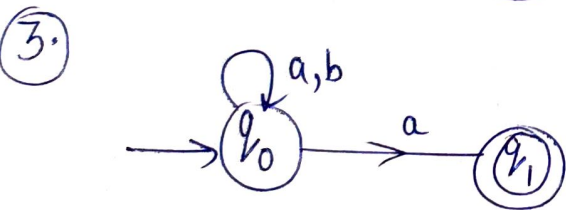
NFA



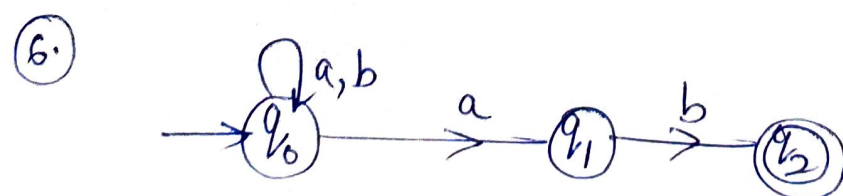
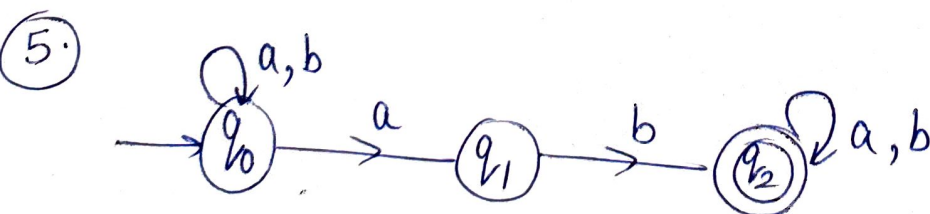
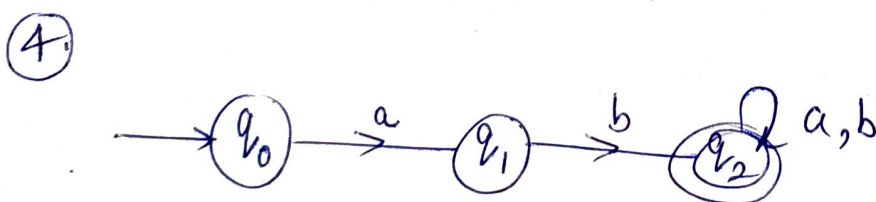
a (a, b's)



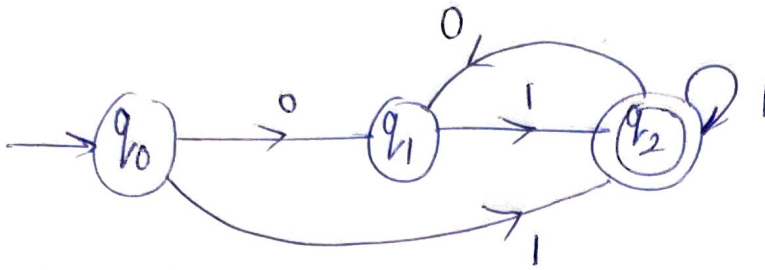
(a's, b's) a (a's, b's)



(a's, b's) a

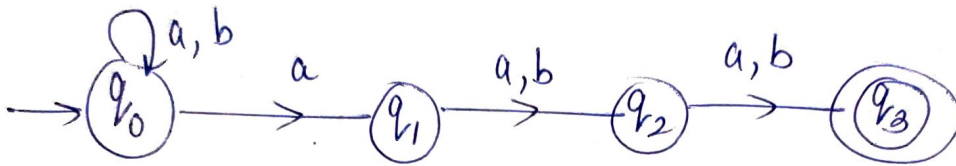


7. 00² 1 → Valid String format

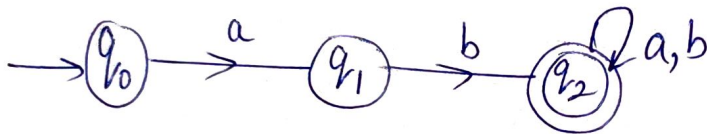


8.

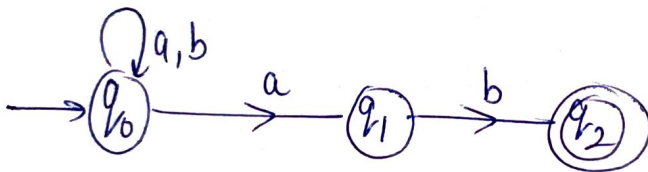
a's, b's a a/b a/b



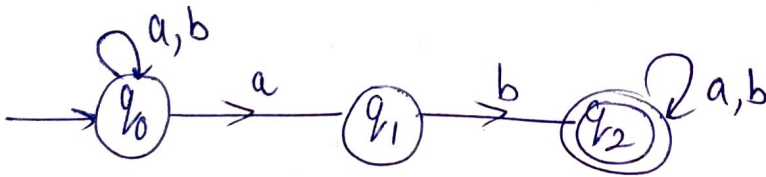
9.



10.



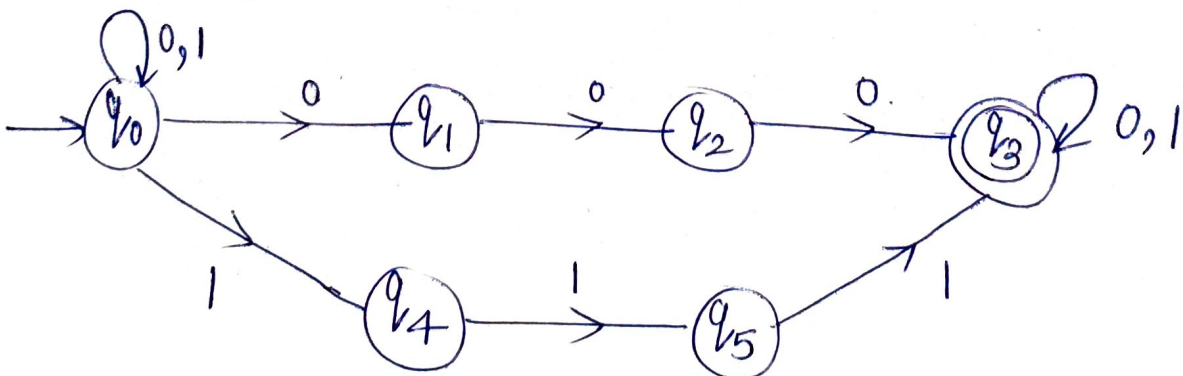
11.



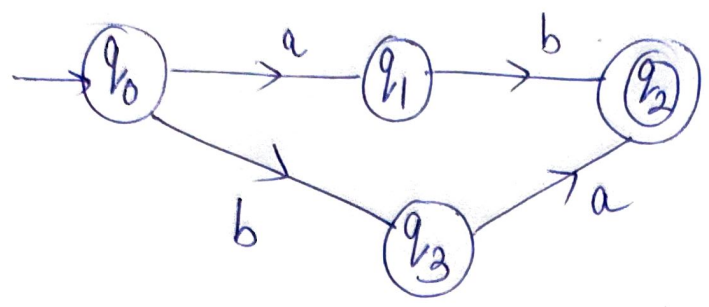
12.

$L = \{ 000, 111, 0000, 0001, \dots \}$

Note: 0²0₁ → contains two consecutive 0's [Valid]



13.



14.

same as Q. 7

15.

same as Q. 8